

Exam Date & Time: 06-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Artificial Intelligence [CSE-4053]

Marks: 50

Duration: 180 mins.

PART-A

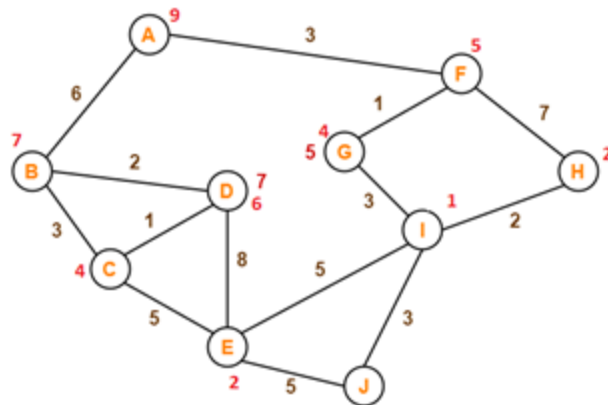
Answer all the questions.

- 1) Analyze the efficiency and optimality of different solution-finding algorithms for the Water Jug problem. (3)
 - A)
 - B) Your AI agent is provided with two water jugs of capacity 5L and 9L respectively. Design the steps for your agent so that it can successfully fill another jug with 6L of water. (3)
 - C) Apply the look-up table and magic square solution approach for tic-tac-toe game. (4)
- 2) State PAGE in Agent Environment and provide PAGE description for Taxi Driver in Bangalore Traffic. (3)
 - A)
 - B) Compare and contrast an intelligent agent architecture considering Goal and Utility based reflex agents with neat Diagram. (3)
 - C) Compare among fully observable and partially observable environments in AI with example. (4)
- 3) Analyse the pros and cons of DFS and BFS for search tree. (3)
 - A)
 - B)

(3)

Apply a cost-effective algorithm for your AI agent to perform the search on the graph given below:

(The numbers written on edges represent the distance between the nodes. The numbers written on nodes represent the heuristic values. Assume A is the starting node and J is the goal node.)



- C) Signify the importance of alpha-beta pruning. Analyze and present suitable example of your choice where tree depth should be ≥ 4 . (4)
- 4) How will the Wumpus World environment be represented in memory? Will it be a grid-based representation, and if so, justify how will the grid be structured? (3)
- A)
- B) An inference algorithm is a computational method or procedure used to derive new information or conclusions from existing knowledge or data. Design an Inference Algorithm for Wumpus world application in AI. (3)
- C) Examine Predicate, Quantifier and FOPL statement for the following. (4)
1. All Dogs Bark
 2. Some cats are Black.
 3. No birds can fly underwater.
 4. Every Monkey eats Bananas.
- 5) Skolemize the followings through proper analysis: (3)
- A)
- (i) $\forall x[\text{Philo}(x) \rightarrow \exists y[\text{Book}(y) \wedge \text{Write}(x,y)]]$
 - (ii) $\exists x \exists y[\text{Philo}(x) \wedge \text{StudentOf}(y,x)]$
- B) Convert the following into clausal form (write down steps): (3)
- $$\exists x \exists y \forall w (\exists z P(f(x, w), y, z) \rightarrow (\exists u Q(x, u) \& \exists v R(y, v)))$$
- C) (4)

Analyze the following cryptarithmic problem and
find the solution:

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BASE
+
BALL
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GAMES
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-----End-----